Practical 2

Linux Commands – Working with Directories and Files

1. `pwd` (Print Working Directory)

Definition

The `pwd` command displays the current working directory.

Use

- To know your current location in the Linux file system.

Syntax

```
pwd
```

Example

```
$ pwd
```

Output

```
/ home / student
```

(Actual output)

```
/home/student
```

2. `cd` (Change Directory)

Definition

Changes the current directory.

Use

- Move from one folder to another.

Syntax

```
cd directory_name
```

Example

```
$ cd Documents
```

Check current location

```
$ pwd
```

Output

```
/home/student/Documents
```

Go Back

```
$ cd ..
```

Output

```
/home/student
```

Go to Home Directory

```
$ cd
```

or

```
$ cd ~
```

Output

```
/home/student
```

Absolute Path

Definition

Complete path starting from the root directory (/).

Example

```
$ cd /home/student/Documents
```

Output

```
/home/student/Documents
```

Relative Path

Definition

Path relative to the current directory.

Current directory

```
/home/student
```

Command

```
$ cd Documents
```

Output

```
/home/student/Documents
```

3. `ls` (List Files)

Definition

Displays files and folders.

Use

- View the contents of a directory.

Syntax

```
ls
```

Example

```
$ ls
```

Output

```
Desktop
Documents
Downloads
Music
Pictures
```

List with Details

```
$ ls -l
```

Output

```
drwxr-xr-x 2 student student 4096 Jun 30 Documents
drwxr-xr-x 2 student student 4096 Jun 30 Downloads
-rw-r--r-- 1 student student   40 Jun 30 notes.txt
```

---

4. `mkdir` (Make Directory)

Definition

Creates a new directory.

Use

- Create folders.

Syntax

```
mkdir folder_name
```

Example

```
$ mkdir LinuxLab
```

Output

(No output)

Check

```
$ ls
```

Output

```
Desktop
Documents
Downloads
LinuxLab
```

---

5. `rmdir` (Remove Directory)

Definition

Deletes an empty directory.

Use

Remove empty folders.

Syntax

```
rmdir folder_name
```

Example

```
$ rmdir LinuxLab
```

Output

(No output)

---

6. `file`

Definition

Shows the type of a file.

Use

Identify whether a file is text, image, executable, etc.

Syntax

```
file filename
```

Example

```
$ file notes.txt
```

Output

```
notes.txt: ASCII text
```

---

7. `touch`

Definition

Creates an empty file.

Use

- Create new files.
- Update timestamp.

Syntax

```
touch filename
```

Example

```
$ touch sample.txt
```

Check

```
$ ls
```

Output

```
sample.txt
```

---

8. `rm`

Definition

Deletes files.

Use

Remove unwanted files.

Syntax

```
rm filename
```

Example

```
$ rm sample.txt
```

Output

(No output)

9. `cp`

Definition

Copies files or directories.

Use

Create duplicate copies.

Syntax

```
cp source destination
```

Example

```
$ cp notes.txt backup.txt
```

Output

(No output)

Verify

```
$ ls
```

Output

```
backup.txt
notes.txt
```

10. `mv`

Definition

Moves or renames files.

Use

- Rename files
- Move files

Syntax

```
mv source destination
```

Example (Rename)

```
$ mv notes.txt notes1.txt
```

Output

(No output)

Check

```
$ ls
```

Output

```
notes1.txt
```

11. `rename`

Definition

Renames one or multiple files.

Syntax

```
rename old_name new_name filename
```

Example

```
$ touch demo.txt
$ rename demo practice demo.txt
```

Output

(No output)

Check

```
$ ls
```

Output

```
practice.txt
```

12. `cat`

Definition

Displays the contents of a file.

Use

- View files
- Create files

Syntax

```
cat filename
```

Example

```
$ cat students.txt
```

Output

```
Rahul
Priya
Amit
Sneha
Neha
```

13. `head`

Definition

Displays the first 10 lines of a file.

Use

View the beginning of large files.

Syntax

```
head filename
```

Example

```
$ head students.txt
```

Output

```
Rahul
Priya
Amit
Sneha
Neha
Ajay
Karan
Pooja
Anita
Ramesh
```

14. `tail`

Definition

Displays the last 10 lines.

Use

View the end of a file.

Syntax

```
tail filename
```

Example

```
$ tail students.txt
```

Output

```
Priya
Amit
Sneha
Neha
Ajay
Karan
Pooja
Anita
Ramesh
Suresh
```

15. `tac`

Definition

Displays a file in reverse order.

Use

Read a file from bottom to top.

Syntax

```
tac filename
```

Example

```
$ tac students.txt
```

Output

```
Suresh
Ramesh
Anita
Pooja
Karan
Ajay
Neha
Sneha
Amit
Priya
Rahul
```

16. `more`

Definition

Displays one screen of a file at a time.

Use

Useful for large files.

Syntax

```
more filename
```

Example

```
$ more students.txt
```

Output

```
Rahul
Priya
Amit
Sneha
Neha
--More--
```

Keys

- Space → Next Page
- Enter → Next Line
- q → Quit

17. `less`

Definition

Displays a file with scrolling support.

Use

Better than `more`.

Syntax

```
less filename
```

Example

```
$ less students.txt
```

Output

```
Rahul
Priya
Amit
...
```

Keys

- ↑ Up
- ↓ Down
- Page Up
- Page Down
- q → Exit

18. `strings`

Definition

Extracts readable text from binary files.

Use

Display printable text.

Syntax

```
strings filename
```

Example

```
$ strings /bin/ls
```

Output

```
GNU
Usage
version
help
Copyright
```

19. `chmod`

Definition

Changes file permissions.

Use

- Give read permission
- Give write permission
- Give execute permission

Syntax

```
chmod permission filename
```

Example

Check permission

```
$ ls -l test.txt
```

Output

```
-rw-r--r-- 1 student student 0 Jun 30 test.txt
```

Give execute permission

```
$ chmod +x test.txt
```

Check again

```
$ ls -l test.txt
```

Output

```
-rwxr-xr-x 1 student student 0 Jun 30 test.txt
```

Practical 3

Linux Commands – Working with Files (System Commands & Compression)

A. Linux System Commands

1. `date`

Definition

The `date` command displays the current system date and time.

Use

- Check the current date and time.
- Display the system clock.

Syntax

```
date
```

Example

```
$ date
```

Sample Output

```
Tue Jun 30 10:45:18 IST 2026
```

2. `cal`

Definition

Displays a calendar.

Use

View the current month's calendar.

Syntax

```
cal
```

Example

```
$ cal
```

Sample Output

```
      June 2026
Su Mo Tu We Th Fr Sa
   1  2  3  4  5  6
7  8  9 10 11 12 13
14 15 16 17 18 19 20
21 22 23 24 25 26 27
28 29 30
```

3. `uptime`

Definition

Displays how long the system has been running.

Use

Check system running time and load.

Syntax

```
uptime
```

Example

```
$ uptime
```

Sample Output

```
10:45:30 up 3 days, 5:20, 2 users, load average: 0.12, 0.08, 0.05
```

---

4. `w`

Definition

Shows who is logged in and what they are doing.

Use

Monitor active users.

Syntax

```
w
```

Example

```
$ w
```

Sample Output

```
USER     TTY      LOGIN@   IDLE   JCPU   PCPU WHAT
student  pts/0    10:00    2:00   0.05s  0.02s bash
```

5. `whoami`

Definition

Displays the current logged-in user.

Use

Know the current username.

Syntax

```
whoami
```

Example

```
$ whoami
```

Sample Output

```
student
```

---

6. `finger`

Definition

Displays information about a user.

Use

View user details (if the `finger` utility is installed).

Syntax

```
finger username
```

Example

```
$ finger student
```

Sample Output

```
Login: student
Name: Student User
Directory: /home/student
Shell: /bin/bash
```

---

7. `uname`

Definition

Displays system information.

Use

Know the operating system and kernel.

Syntax

```
uname
```

Example

```
$ uname
```

Output

```
Linux
```

Detailed Information

```
$ uname -a
```

Sample Output

```
Linux ubuntu 6.8.0 x86_64 GNU/Linux
```

---

8. `man`

Definition

Displays the manual page of a command.

Use

Learn how a command works.

Syntax

```
man command_name
```

Example

```
$ man ls
```

Output

```
LS(1)
NAME
ls - list directory contents
```

Press q to quit.

9. `df`

Definition

Displays disk space usage.

Use

Check free and used disk space.

Syntax

```
df -h
```

Example

```
$ df -h
```

Sample Output

```
Filesystem      Size Used Avail Use%
/dev/sda1        50G  20G   28G  42%
```

---

10. `du`

Definition

Displays directory size.

Use

Know how much space a folder occupies.

Syntax

```
du -h folder_name
```

Example

```
$ du -h Documents
```

Sample Output

11. `free`

Definition

Displays memory usage.

Use

Check RAM usage.

Syntax

```
free -h
```

Example

```
$ free -h
```

Sample Output

```
              total   used   free
Mem:          7.6G   2.4G   5.2G
Swap:         2.0G     0G   2.0G
```

12. `whereis`

Definition

Finds the location of a command and its manual pages.

Use

Locate binaries and documentation.

Syntax

```
whereis command
```

Example

```
$ whereis ls
```

Sample Output

```
ls: /usr/bin/ls /usr/share/man/man1/ls.1.gz
```

13. `which`

Definition

Displays the executable path of a command.

Use

Find the command location.

Syntax

```
which command
```

Example

```
$ which python3
```

Sample Output

```
/usr/bin/python3
```

14. `grep`

Definition

Searches for text inside a file.

Use

Find words or patterns.

Syntax

```
grep "word" filename
```

Example

Suppose `students.txt` contains:

```
Rahul
Priya
Amit
Sneha
```

Command

```
$ grep "Amit" students.txt
```

Output

```
Amit
```

15. `locate`

Definition

Searches files quickly using a database.

Use

Locate files anywhere in the system.

Syntax

```
locate filename
```

Example

```
$ locate notes.txt
```

Sample Output

```
/home/student/Documents/notes.txt
```

16. `find`

Definition

Searches files in directories.

Use

Find files by name.

Syntax

```
find path -name filename
```

Example

```
$ find /home/student -name notes.txt
```

Sample Output

```
/home/student/Documents/notes.txt
```

B. Compression Commands

17. `tar`

Definition

Creates or extracts archive files.

Use

Combine multiple files into one archive.

Syntax

Create Archive

```
tar -cvf archive.tar foldername
```

Extract Archive

```
tar -xvf archive.tar
```

Example

Create archive

```
$ tar -cvf LinuxLab.tar LinuxLab
```

Sample Output

```
LinuxLab/
LinuxLab/file1.txt
LinuxLab/file2.txt
```

Extract archive

```
$ tar -xvf LinuxLab.tar
```

Sample Output

```
LinuxLab/
LinuxLab/file1.txt
LinuxLab/file2.txt
```

---

18. `gzip`

Definition

Compresses a file to reduce its size.

Use

Save storage space.

Syntax

```
gzip filename
```

Example

```
$ gzip notes.txt
```

Sample Output

(No output)

Check the file

```
$ ls
```

Output

```
notes.txt.gz
```

Decompress

```
$ gunzip notes.txt.gz
```

Output

(No output)

Verify

```
$ ls
```

Output

```
notes.txt
```